

Part I — The committer on its own

Setup: the committer machines

Nineteen machines, each 64 cores and 156 GiB RAM, running three signature verifiers, one coordinator, one sidecar, one load generator, six validator-committers, twelve YugabyteDB tablet servers and three YugabyteDB masters. The six validator-committers and the three masters sit on nine of the twelve database machines, and no machine runs both a validator-committer and a master.

Storage is two 969 GiB XFS volumes per machine, `noatime`, mounted `/data1` and `/data2`. Each database node passes YugabyteDB both volumes as separate data directories (`--fs_data_dirs`, one path per volume) rather than a stripe. The sidecar's ledger uses `/data1` alone.

Every transaction slot takes a fresh key, so there are no read-write conflicts except in the panel that introduces them deliberately. Blocks come from the load generator's embedded mock ordering service, which cuts and signs them itself and serves them to the sidecar over the Atomic Broadcast API, so no ordering service is in the path.

Transaction signatures satisfy namespace 0's endorsement policy, a threshold rule over the endorsing user's signing certificate. The load generator creates the namespace itself on this arm, registering the policy over a key pair it holds, so the policy and the key it signs with agree by construction and the scheme is a free choice; these figures use Ed25519.¹

A measured point is the highest offered rate that arrived in full, committed within 2%, kept p99 under 1s and held a flat in-flight count, confirmed by a 300s hold on a fresh deployment.

Why this setup

The shape is not the paper's. The published committer deployment is nine validator-committers on nine database nodes, dual 48-core servers with 64 GB. What follows therefore compares deployments, not matched hardware.

Both volumes, not a stripe, so YugabyteDB sees the disk boundary and places data across it itself. The data directories are siblings of the deployment directory rather than children of it, so the two are wiped independently.

A conflict-free workload measures the pipeline's ceiling rather than its conflict behaviour, at the cost of a state database and a ledger that grow with every transaction. Conflicts are then introduced deliberately as their own variable.

Queue flatness is part of the definition because a rate can be committed in full out of a growing queue, which reports the queue's drain time as latency.

¹The scheme is not a variable here: against Ed25519's 605,000, 518,000, 419,000 and 274,000 tps at one to four read-write operations, ECDSA holds 653,000, 518,000, 389,000 and 275,000 — mean ratio 1.003, scattered 8% in *both* directions and every point inside the 8.8% repeatability established below. Both series are confirmed holds. The panel's shape survives too, throughput falling 58% from one read-write operation to four against Ed25519's 55%, so the per-key mechanism argued below belongs to the pipeline and not to the signature scheme.

Throughput and tail latency

Throughput falls with transaction size (Fig. 1), 604,545 tps at one read-write operation to 274,364 at four. Comparing that with the published figure needs the axes aligned: the paper's x axis counts inputs and outputs, and one in with one out is a *two* read-write total, so its N/N sits at our $2N$. Only two of our four sizes therefore have a published counterpart. At two read-writes this cluster holds 518,000 tps against the paper's 474,000, 9% ahead; at four it holds 274,364 against 419,000, 35% behind. Over those same two sizes our throughput falls 47% where the paper's falls 12%, so whatever this pipeline pays per key, it pays much more steeply than the published one. The paper's six- and eight-key points were never measured here, and our one- and three-key points have nothing to compare against.

The mechanism behind that slope is not the published one, which is lock contention in the coordinator's dependency graph. Here both lock-wait histograms have a count rate of exactly zero, and the graph's cost is per key instead — about 420 ns each, flat from one key per transaction to eight.

Invalid signatures are close to free: 518,000 tps with none, then 517,455, 519,455 and 518,727 at 10%, 20% and 30% — all four flat within 0.4%, far inside this cluster's 8.8% repeatability, so rejection costs nothing measurable in throughput. It is not the published 10% *gain* either. Where it does cost is the tail. Driven at a common 559,872 tps, above the qualifying rate, every share still delivers that rate to within 2.4%, but p99 goes from 0.70s with nothing rejected to 1.2s to 3.0s once a tenth or more is: rejection is cheap in throughput and expensive in latency.

Double spends do not reproduce, and a deployment setting is why. At the 120-way tablet pre-split this cluster uses, 0% holds 518,000 tps and 5% collapses: no offered rate qualified. Its best attempt at the bound delivered 8,160 tps with a 9.9s tail, and its highest-throughput attempt 21,636 tps with the tail pinned at the 60s measurement ceiling — the former with the busiest machine at 18% CPU, so neither capacity nor a queue. 5% is the only conflict share measured at this split; the higher shares were measured at the default split instead.

At the default split the same shapes hold: 41,273, 30,000 and 40,909 tps at 10%, 20% and 30% with 334 ms to 533 ms p99, flat rather than declining — while costing the conflict-free case by a factor this experiment cannot pin down: at the default split *no* rate between 155,204 and 294,780 tps got a confirmed hold with no conflicts at all, so Fig. 1 shows no bar there. The split is therefore a workload-dependent trade rather than a tuning win, and why 120 tablets collapse under conflicts is unresolved.

Latency against throughput

Both curves carry two read-write operations over 32 B keys, 262 B serialized, and differ only in the number of transactions the generator puts in a block.

At 10,000-transaction blocks the median has a floor near 125 ms that does not fall with the rate, since nothing in a block moves until the block is cut (Fig. 2). At 500 transactions a block the floor is 40 ms, and 380,100 tps holds at

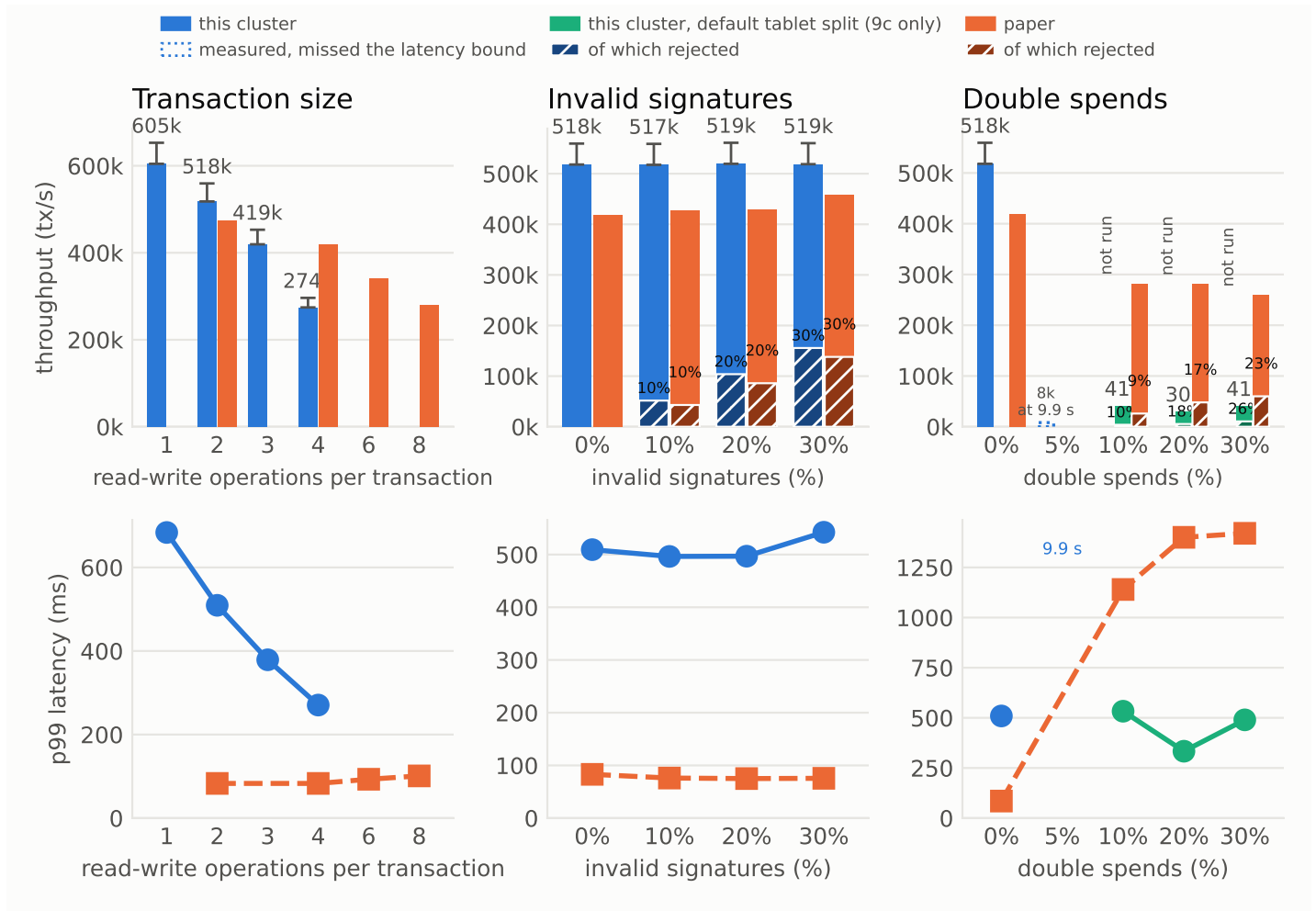


Figure 1: Committer throughput (top) and 99th-percentile latency (bottom) against transaction size, invalid-signature share and double-spend share, beside the published Figure 9. Throughput counts transactions finished, committed plus rejected, and the hatched part of a bar is the rejected share with its percentage. Whiskers are the search’s 8% step, one-sided. A dotted outline is a condition that was measured and missed the latency bound, labelled with the tail that disqualified it; *not run* marks one never attempted at that configuration.

125ms median and 208ms p99 — the median large blocks give at 10,000 tps, at thirty-eight times the rate. The top of that curve measures the generator, whose block preparation caps it near 850 blocks a second (425,000 tps at this size; 450,000 offered delivered 432,309), so the committer’s own ceiling at 500 transactions is not resolved here.

Part II — End to end, with a real ordering service

The second arm puts a real Arma ordering service in the path. The committer tier is **unchanged from Part I** — the same nineteen machines, the same components, the same tuning — so a difference between the two arms is attributable to ordering.

Setup: the ordering machines

Twenty further machines, each 32 cores and 78 GiB, holding four parties of a router, a consenter, an assembler and one batcher per shard. Routers, consenters and assemblers take a

machine each, twelve in all; the batchers share the remaining eight. Two configurations are compared:

- **Four shards.** Sixteen batchers, two to a machine, each with a volume to itself (/data1 and /data2).
- **Eight shards.** Thirty-two batchers, four to a machine, two sharing each volume.

Storage is two 485 GiB XFS volumes per machine, mounted as on the committer machines. Routers, consenters and assemblers write to /data1; the batchers are distributed across both volumes as above.

A batcher cuts batches and the assemblers turn each one into a block, so on this arm the block-size knob is the shared configuration’s `Batching.BatchSize.MaxMessageCount` rather than a load-generator setting, and `BatchCreationTimeout` is 500ms. Signatures are ECDSA P-256, and here the scheme is forced rather than a free choice. `fxconfig` registers namespace 0 with a threshold rule over the endorsing user’s MSP signing certificate, and only that key satisfies it — every other scheme signs with a synthetic key pair no

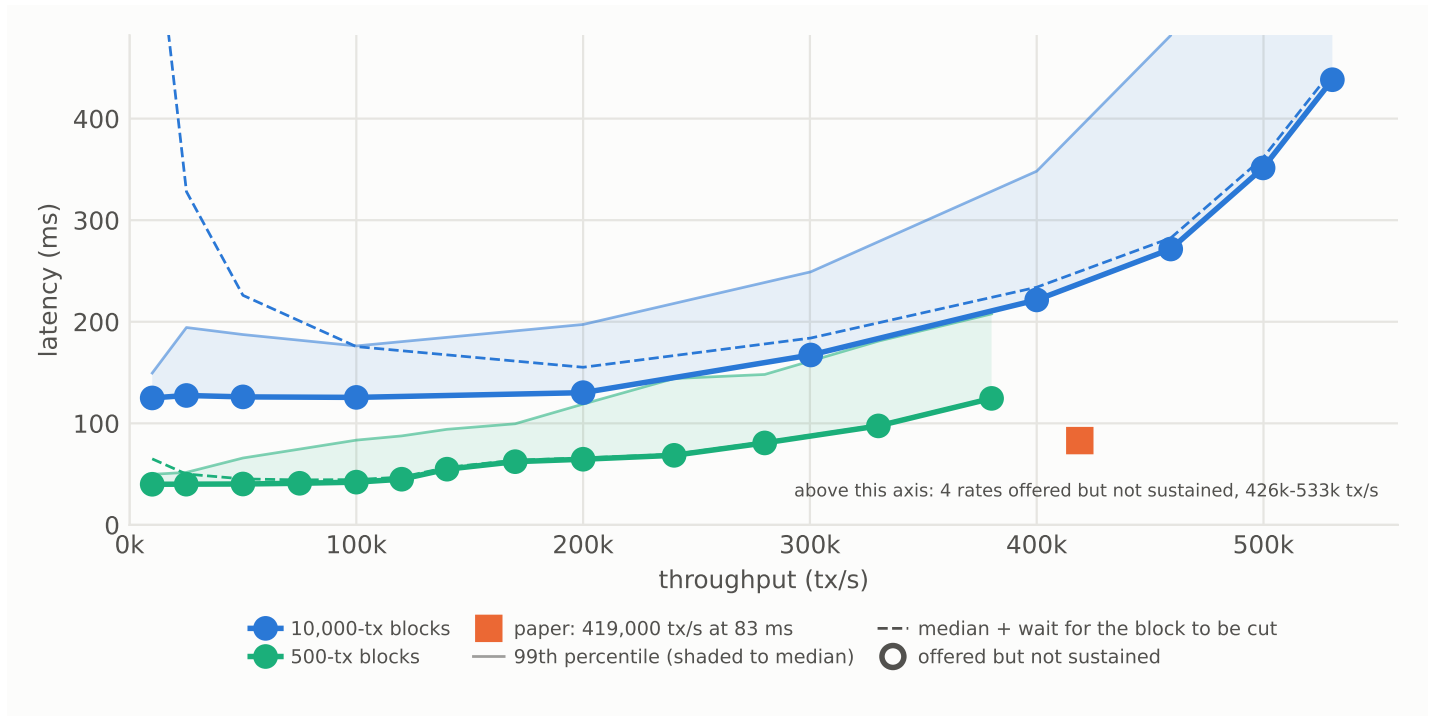


Figure 2: Latency against throughput for the committer alone, at two block sizes and a fixed workload of two read–write operations. Thick line: the median; the thin line above it the 99th percentile, shaded between; dashed: the median plus the mean wait for a block to be cut, which the measurement excludes. Hollow markers are rates offered but not sustained. The latency axis is floored and then grown to contain every sustained rate; rates above it are summarised in a corner.

policy in this deployment knows, so the workload would be rejected wholesale for invalid signatures. Namespace creation is governed by a different policy again: an MSP rule on the `_meta` namespace, which the load generator has no material for, so `fxconfig` creates the namespace before load is applied.

Why this setup

Two batchers to a machine is measured rather than assumed. At two shards the service capped near 158,000 tps *per shard* whatever the batch size or decision interval, while batcher processes sat at 15% to 20% of a 32-core machine — so the constraint is per shard and the spare capacity for a second was already on the box.

Both shard counts are measured because that reading predicts more shards buy throughput, and the two configurations separate the two things a shard needs: a core budget and a disk. Four shards give each co-located batcher its own volume; eight give twice the shards but make two share one.

These machines are half a committer machine — half the cores, half the memory and, per Table 2, half the disk bandwidth — so a ceiling measured on this arm belongs to these machines before it belongs to ordering.

Latency against throughput, by shard count and storage

Both ladders here carry the workload of Fig. 2 — two read–write operations over 32 B keys, 262 B serialized, every slot taking a fresh key — so the two arms are comparable at

matched rates and differ only in whether an ordering service is in the path.

End to end, ordering included, this deployment sustains 499,091 tps at 658 ms p99, with every offered rate up to that arriving in full and no aborted transactions at any rung. Two comparisons place it. The published work has no end-to-end figure, but its ordering service alone reaches 430,000 tps at four parties and four shards — so the complete pipeline here exceeds the published figure for one part of it. And the same committer at the same workload without an ordering service holds 518,000 tps, reaching 530,364 on the ladder of Fig. 2, so putting real ordering in the path costs under 6% of the committer’s own ceiling — inside this cluster’s repeatability rather than a separate order of magnitude.

Both configurations sustain the same throughput and hit the same knee: 499,091 tps at four shards against 500,000 at eight, 0.2% apart, and neither sustains 550,000 (Fig. 3). Four shards is the better configuration on every other axis — p99 lower at every matched rate, by 44 ms to 248 ms (658 ms against 754 ms at the knee), fewer processes, and a volume per batcher rather than two sharing one.

Doubling the shards buys nothing because ordering is not what limits this arm. Below 400,000 tps the busiest machine is a router; at the top of both ladders it is a validator–committer at 76% to 82%. Extra shards add capacity to the tier that is not binding.

The knee itself cannot be attributed to a single tier. At 500,000 tps no machine anywhere exceeds 82%: the busiest validator–committer is at 81% and *the load generator is at 77%*, four points behind it. So the ceiling is neither a CPU wall on any tier nor separable by this experiment — the two

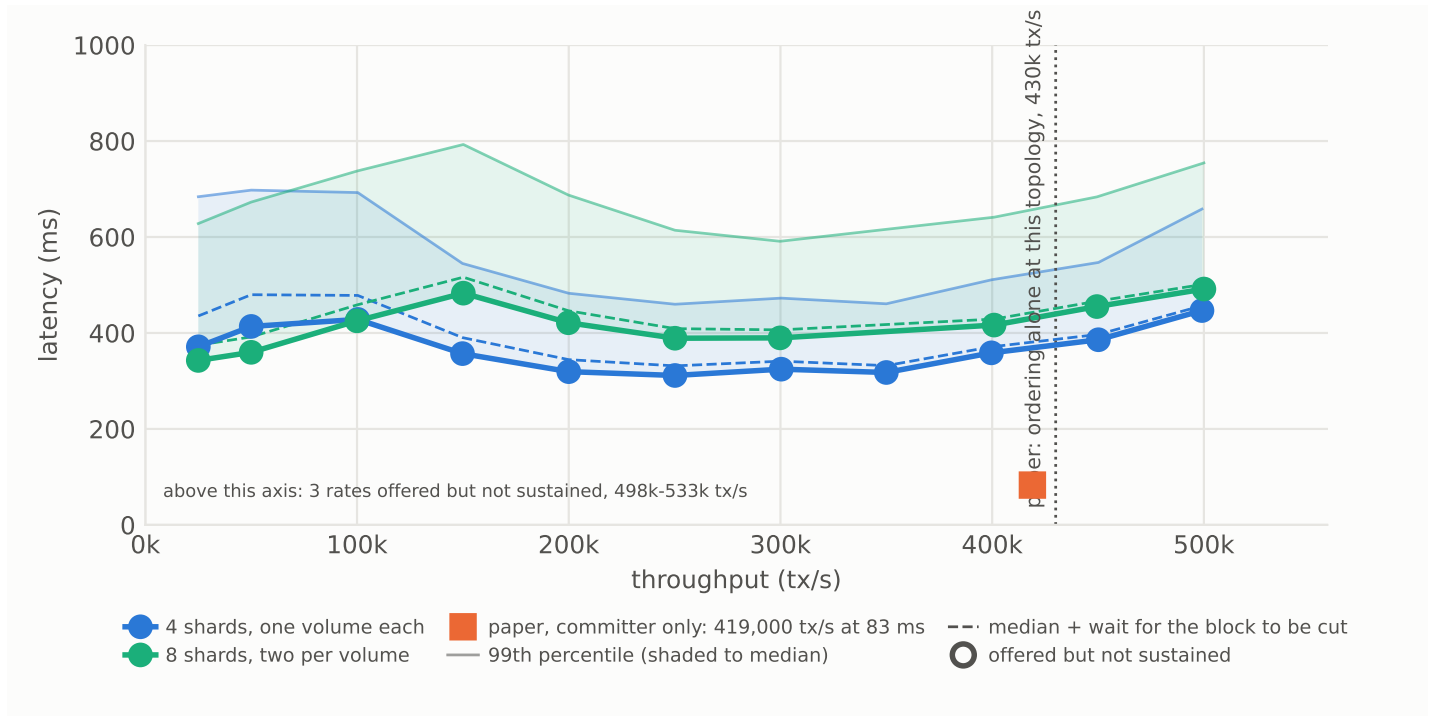


Figure 3: Latency against throughput end to end, with a real ordering service in the path, by shard count and batcher storage layout. Marks as Fig. 2. The dotted vertical line is the published ordering-alone figure for this topology.

candidates are within each other’s margin, and a ladder on one deployment cannot tell them apart. Naming the busiest host names a correlate, not a cause.

Extra shards also cost latency, for a reason the block size makes plain. At a given total rate each of eight shards sees half the per-shard rate that four do, so batches take twice as long to reach 10,000 transactions and the 500ms `BatchCreationTimeout` cuts them early over twice as wide a range of rates. The eight-shard curve shows exactly that shape: p99 rises to 793ms at 150,000 tps, then falls back as the rate grows enough to fill a batch before the timer fires — 687, 614 and 591 ms at 200,000, 250,000 and 300,000 tps.

Latency against throughput, by batch size

Batch size is the latency–throughput knob here, and it trades one for the other almost exactly (Fig. 4), at the same workload again. Cutting the batch from 10,000 transactions to 500 removes most of the latency at every rate the two share — p99 197ms against 693ms at 100,000 tps, 286ms against 460ms at 250,000, and mean latency between a quarter and a half of what large batches give. That is the predicted effect and its direction confirms the mechanism: a batch a twentieth the size fills twenty times sooner, so the 500ms `BatchCreationTimeout` stops being what cuts it.

What it costs is the ceiling, and the reason is exact. The block rate tracks the offered rate precisely — 200 blocks a second at 100,000 tps, 400 at 200,000, 500 at 250,000 — and between 500 and 600 blocks a second the arm stops working. 250,000 tps is therefore not a transaction limit at all: it is 500×500 , a *block*-rate ceiling near 500 a second. At 10,000 transactions a batch the same block rate would carry five million transactions a second, which is why that configuration

is limited by transactions instead and reaches 499,091.

So the two configurations are bounded by different things, and the choice is a real trade rather than a tuning win: small batches halve the latency and halve the ceiling.

Throughput against transaction size

Throughput falls with transaction size, but far more slowly than in the published sweep, and the byte rate is what shows why (Fig. 5). The sweep keeps the same two read–write operations and varies only the value each one carries, so the serialized envelope is 202 bytes plus twice that value. The published measurement is bandwidth-bound past 300 B: each of its points sits within 10% of 120 MB s^{-1} , so its rate is simply bytes per second divided by size. This deployment is not bandwidth-bound anywhere in the range. Its byte rate *rises* with size, from 122 MB s^{-1} at 300 B to 410 MB s^{-1} at 4 KiB, because what it pays per transaction dominates what it pays per byte: halving the transaction count at twice the size is a net gain. Throughput therefore falls about $0.76\times$ per doubling rather than the $0.5\times$ a bandwidth-bound service gives.

The tabulated 512 B figure is nominally above the 300 B one, which is not a real inversion: the 300 B point’s two holds span 408,000 to 443,800 tps and 512 B sits inside that, so the sweep is flat between those sizes rather than rising. It is the kind of difference the repeatability figure in Table 1 exists to rule out.

Where that stops is measurable, and it is the assembler’s disk. At 4 KiB the arm delivers 113,000 to 123,000 tps whether it is offered 118,000 or 163,200, so throughput plateaus near 120,000 tps — 492 MB s^{-1} of envelope bytes — with no machine anywhere above 54% CPU, so it is not a processing limit. The highest rate that also meets the la-

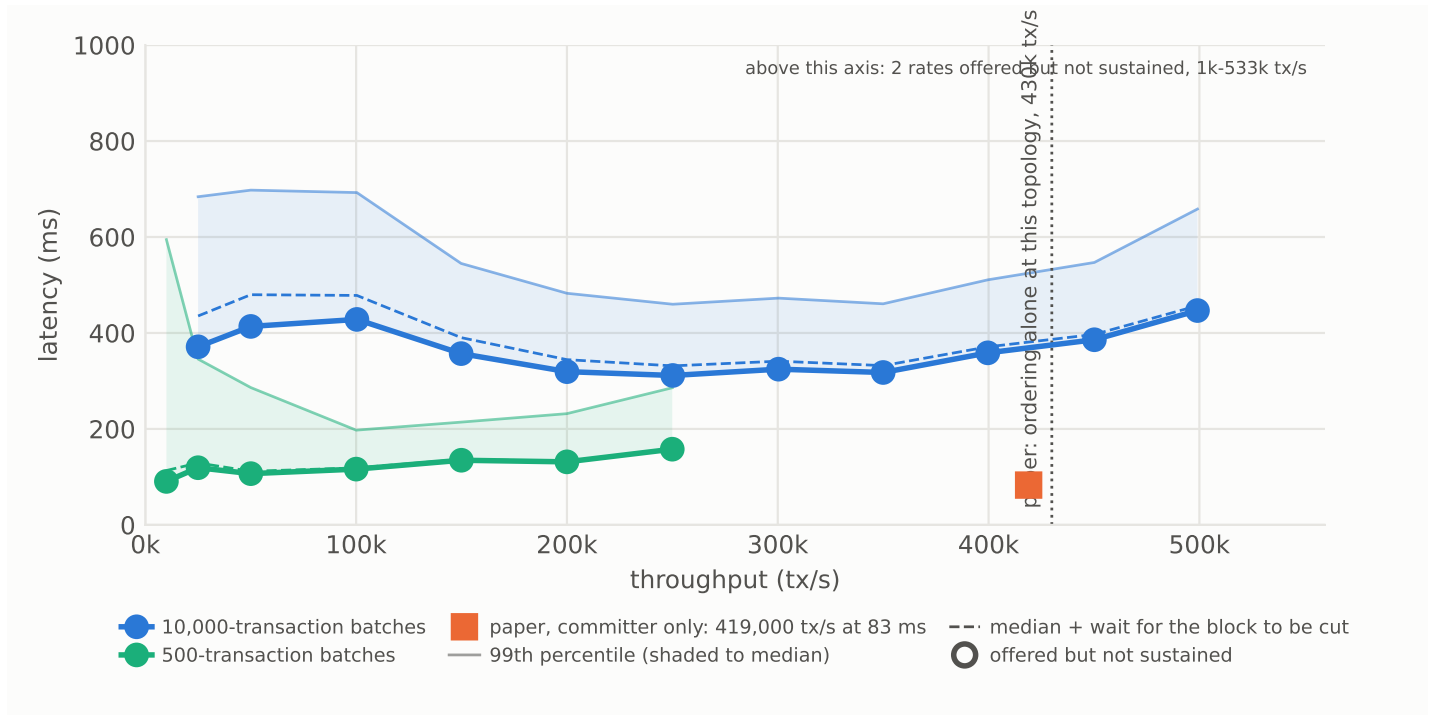


Figure 4: Latency against throughput end to end, by batch size, on four shards with a volume per batcher. Marks as Fig. 2.

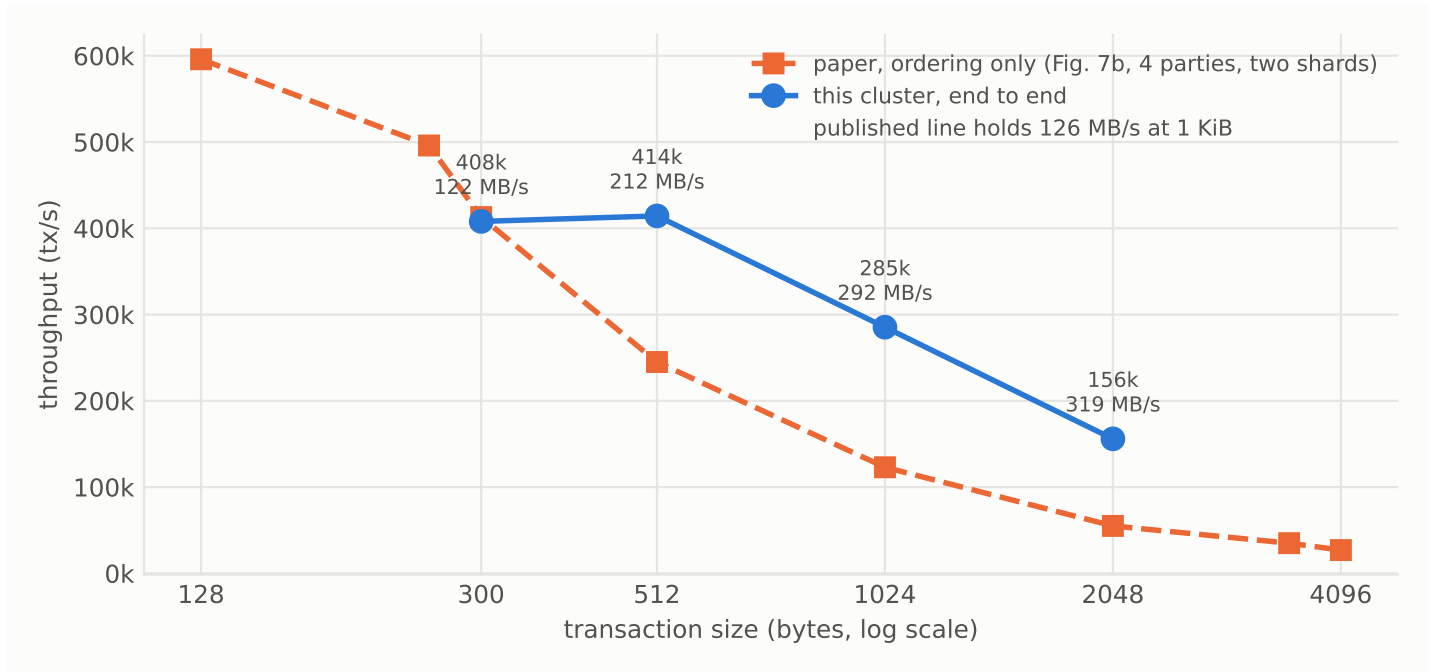


Figure 5: Throughput against transaction size end to end, beside the published Figure 7b. Each point is annotated with the byte rate it implies.

tency bound, which is what Table 1 reports, is lower still at 100,000 tps. The assemblers at that point are writing 527 MB s^{-1} to 546 MB s^{-1} to their data volume, against the 529 MiB/s sequential write Table 2 measures on these machines: the device is saturated. Each of the four assemblers writes the *complete* block stream, where a batcher writes only its own shard's share, so the assembler is where a byte rate first meets a disk. At 2 KiB the same figure is 319 MB s^{-1} , three fifths of that ceiling, which is why that point holds cleanly and this one does not.

So the sweep has two regimes rather than one. Below 2 KiB the arm is limited per transaction and the byte rate climbs; at 4 KiB it is limited per byte by one component's disk, and the published bandwidth-bound model finally applies here too, at about four times the bandwidth.

The consequence for the comparison is that only the direction is shared. Both lines fall, and this one is above the published line at every size they have in common — but it is a whole pipeline against one of its parts, at four shards against two, so the gap is not a speedup and the shapes are

Table 1: The size sweep end to end, with the evidence class of each point, since they do not all meet the same bar. A hold is tabulated in preference to a higher probe wherever both exist. At 1 KiB and above a hold does not fit in one deployment: each assembler writes the complete block stream, so a 485 GB volume fills in under twenty minutes at these byte rates, less than a rate search followed by a 300s hold. The 300B point is the lower of two holds 8.8 % apart, so all figures are given to three significant figures.

Size	Throughput (tx/s)	Byte rate	Evidence
300 B	408,000	122 MB/s	two holds, lower one
512 B	414,000	212 MB/s	confirmed hold
1 KiB	285,000	292 MB/s	probe; volume filled
2 KiB	156,000	319 MB/s	confirmed hold
4 KiB	100,000	410 MB/s	probe; disk-bound

Table 2: **fiio 3.35**, **libaio**, **direct=1**, 20s per pattern, run on a volume the deployment was not using at the time. Committer figures are the range across a database node and the sidecar. A synchronous engine caps fio’s queue depth at 1 while still reporting the depth asked for, so the deep rows use **libaio**.

Pattern	Depth	Committer	Ordering
1 MiB seq. write	4	1,058 MiB/s	529 MiB/s
4 KiB O_DSYNC write	1	2,400 IOPS	3,249 IOPS
8 KiB rand. read	32	126,000 IOPS	67,689 IOPS
8 KiB rand. write	32	102,000 IOPS	51,165 IOPS

not measuring the same limit.

Storage characterisation

Two properties of these volumes bound the results above, and both had to be measured: the volumes are **virtio** devices reporting neither a model nor a meaningful rotational flag, so the backing media cannot be identified from inside the guest. Table 2 gives **fiio 3.35** in the four patterns this deployment produces.

First, ordering machines have half the disk bandwidth as well as half the cores, at almost exactly a factor of two on every bandwidth row, at round figures repeating to four significant figures across machines and volumes — these are provisioned caps, not device characteristics, so a ceiling on that arm must be read against the cap before it is attributed to ordering. The synchronous pattern is the exception, where the ordering volumes are the faster of the two (297 μ s against 395 μ s at the median). Second, durability is the expensive direction: making one 4 KiB record durable with an **O_DSYNC** write costs about 400 μ s, capping any per-record design near 2,400 operations a second, three orders of magnitude below the rates measured here. The ledger survives that only by batching — it appends a whole block and syncs every hundredth. Syncing alone is cheap by comparison, an **fdatasync** after a buffered write returning in about 0.08ms: it is the write, not the barrier, that costs.

Threats to validity

The load generator is the instrument, and in one figure it is the limit: its block preparation caps the small-block curve in Fig. 2. Elsewhere it is close to being one. At the end-to-

end knee it runs at 77 % of its 64 cores against the busiest validator–committer’s 81 %, and on the 300B size point at 80 % against 81 % — tied, within the noise. Every figure near 500,000 tps should therefore be read as a bound on the pair, generator and pipeline together, not on the pipeline alone. Separating them needs a second generator, and there is no spare machine for one without taking it from the committer. A knee is resolved to the search’s 8 % step, so it is a lower bound, drawn one-sided in Fig. 1. The baseline drifts 10 % between days, so only same-day comparisons are used, and within a day repeatability is no better: one configuration measured twice gave holds 8.8 % apart, and two configurations that are the same workload reached by separate searches differed by 7.9 %. A difference under about a tenth is therefore not a result here, and no claim in this section rests on one.

A ladder climbs on a single deployment, so rate and table fill rise together along it. Repeating the rates around the knee on fresh deployments moves the boundary by less than the spread above, so the knee is a property of the rate; the tail near it does grow with fill. Every rung records the fill it ran against.